PriviScan

Setup & Run Manual

For users who clone the complete project from GitHub and need to run it for the first time.

The source code is already available in the GitHub repository. This manual focuses only on installing the required software, downloading the project, preparing the environment, running the project, and fixing common setup problems.

Quick path

```
Install Software
      ↓
Clone PriviScan from GitHub
      ↓
Open PriviScan folder
      ↓
Create .venv
      ↓
Activate .venv
      ↓
Install requirements.txt
      ↓
Run a test
      ↓
Start PriviScan
      ↓
Open in browser
```

1. Before You Start

You need a Windows computer with internet access and permission to install software.

Required software

• Windows 10 or Windows 11

• Python 3.11.x

• Git for Windows

• Visual Studio Code (recommended, but not required for running)

• Internet connection for package/model downloads

You do **not** need to manually copy the PriviScan source files. The complete project is obtained from GitHub.

2. Install Python

Install Python 3.11.x. During installation, enable **Add Python to PATH**.

Open PowerShell and check:

```
python --version
py -3.11 --version
```

You should see Python 3.11.x.

3. Install Git

Install Git for Windows, then verify:

```
git --version
```

4. Install VS Code

VS Code is recommended for opening the project, viewing files, and using the integrated terminal. After installing it, you can open the PriviScan folder through **File** → **Open Folder**.

5. Get PriviScan from GitHub

Open PowerShell. Go to the folder where you want to keep the project.

```
cd "C:\Users\<YOUR_USERNAME>\OneDrive\Documents\KSU\SEM 6\Capstone Design_1"
```

Clone the repository. Replace the placeholder with the actual PriviScan GitHub repository URL.

```
git clone <YOUR_GITHUB_REPOSITORY_URL> PriviScan
cd PriviScan
```

Verify that you are inside the project:

```
dir
git status
```

If you already cloned the repository previously, do not run git clone again. Instead:

```
cd PriviScan
git pull
```

6. Create the Python Virtual Environment

The virtual environment keeps PriviScan's Python packages separate from other projects.

```
py -3.11 -m venv .venv
```

Activate it:

```
.\.venv\Scripts\Activate.ps1
```

After activation, the terminal should show **(.venv)** at the beginning of the prompt.

Verify:

```
python --version
python -c "import sys; print(sys.executable)
```

If PowerShell refuses to activate the environment:

```
Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope CurrentUser
.\.venv\Scripts\Activate.ps1
```

7. Install PriviScan Dependencies

Make sure **(.venv)** is active, then run:

```
python -m pip install --upgrade pip setuptools wheel
python -m pip install -r backend\requirements.txt
```

This installs the Python packages required by the version of PriviScan stored in GitHub. Do not manually install individual packages unless an error specifically requires it.

The first OCR/AI run may also download or initialize model files. This can take longer than later runs.

8. Project File Tree — Reference Only

The code is already in GitHub. This tree is only to help users recognize the project after cloning it.

```
PriviScan/
■
■■■ .venv/
■
■■■ backend/
■   ■■■ __init__.py
■   ■■■ main.py
■   ■■■ analyze.py
■   ■■■ vision.py
■   ■■■ vision_evaluation.py
■   ■■■ evaluation.py
■   ■■■ protection.py
■   ■■■ requirements.txt
■   ■
■   ■■■ ocr/
■   ■   ■■■ __init__.py
■   ■   ■■■ detector.py
■   ■
■   ■■■ detection/
■   ■   ■■■ __init__.py
■   ■   ■■■ ai_detector.py
■   ■   ■■■ sensitive.py
■   ■   ■■■ context.py
■   ■
■   ■■■ risk/
■       ■■■ __init__.py
■       ■■■ scoring.py
■
■■■ dataset/
■   ■■■ mixed/
■   ■■■ week7_vision/
■
■■■ uploads/
■
■■■ evaluation_results.json
■■■ vision_evaluation_results.json
```

9. First Setup Test

Before starting the web application, test the main project pipeline.

Test OCR:

```
python -m backend.ocr.detector dataset\mixed\info.png
```

Test the complete analysis:

```
python -m backend.analyze dataset\mixed\info.png
```

If these commands run successfully, the main Python environment and project dependencies are working.

10. Start PriviScan

Start the application from the PriviScan root with the virtual environment active:

```
python -m backend.main
```

If the application starts successfully, open the local address shown in the terminal. Normally it is:

```
http://127.0.0.1:8000
```

Alternative startup command:

```
python -m uvicorn backend.main:app --reload
```

Stop the application with **Ctrl+C**.

11. Basic Usage

```
1. Open PriviScan in the browser.
2. Upload an image.
3. Run analysis.
4. Review the detected sensitive information.
5. Review the risk result.
6. Choose the protection option.
7. Generate the protected image.
8. Check the protected output.
```

The exact interface comes from the version of the project stored in GitHub.

12. When the GitHub Project Is Updated

If you already have PriviScan installed and want the newest version:

```
cd "C:\path■o\PriviScan"
.\.venv\Scripts\Activate.ps1
git pull
python -m pip install -r backend\requirements.txt
```

Then run the normal test/start commands again.

13. Git Commands — Quick Reference

Check status:

```
git status
```

Download the newest repository changes:

```
git pull
```

See recent commits:

```
git log --oneline -10
```

If you are a contributor and need to upload your changes:

```
git add .
git commit -m "Update PriviScan"
git push
```

If your changes are already committed, **git push** is enough.

14. Common Setup Problems

Problem: python is not recognized

Reinstall Python and enable Add Python to PATH, then reopen PowerShell.

Problem: python -m backend.analyze says No module named backend

You are probably not in the PriviScan root folder. Run:

```
cd "C:\path\to\PriviScan"
.\\.venv\\Scripts\\Activate.ps1
python -m backend.analyze dataset\\mixed\\info.png
```

Problem: requirements installation fails

```
python -m pip install --upgrade pip setuptools wheel
python -m pip install -r backend\requirements.txt
```

If it still fails, keep the complete terminal error; dependency errors can depend on the Python version and machine.

Problem: PowerShell blocks .venv activation

```
Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope CurrentUser
.\.venv\Scripts\Activate.ps1
```

Problem: Port 8000 is already in use

```
python -m uvicorn backend.main:app --reload --port 8001
```

Then open http://127.0.0.1:8001.

Problem: First OCR/AI run takes a long time

The first run may initialize/download model files. Wait for it to finish. Later runs should reuse the downloaded models.

15. Complete Setup Checklist

```
[ ] Python 3.11 installed
[ ] Git installed
[ ] VS Code installed (recommended)
[ ] PriviScan cloned from GitHub
[ ] Terminal is inside PriviScan
[ ] .venv created
[ ] .venv activated
[ ] requirements.txt installed
[ ] OCR test works
[ ] Analysis test works
[ ] PriviScan server starts
[ ] Browser opens the application
```

16. One-Block Setup Command List

For a completely new computer, after installing Python, Git and VS Code:

```
git clone <YOUR_GITHUB_REPOSITORY_URL> PriviScan
cd PriviScan

py -3.11 -m venv .venv
.\\.venv\\Scripts\\Activate.ps1

python -m pip install --upgrade pip setuptools wheel
python -m pip install -r backend\\requirements.txt

python -m backend.ocr.detector dataset\\mixed\\info.png
python -m backend.analyze dataset\\mixed\\info.png

python -m backend.main
```

Important

This PDF is a setup/run guide only. The complete PriviScan source code, requirements file, datasets, and project files should be obtained from the project's GitHub repository. If the GitHub project changes, follow the latest repository files and update this guide if necessary.